

Nicola Muca Cirone

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EDUCATION	PhD Mathematics - CDT Mathematics of Random Systems, Imperial College London - University of Oxford 2022-2025 • Deep Learning, Mathematical Finance, Rough Paths Analysis. • Supervised by <i>Dr. Cristopher Salvi</i>. • Researching scaling limits of Neural Networks • Working in theoretical foundations for State Space Models. • Developing new Kernel Methods for Time Series analysis. MSc Mathematics, University of Pisa - ETH Zürich 2021-2022 • Honour's degree (110/110 cum Laude) GPA 30 e Lode • Stochastic Calculus, Stochastic PDEs, Mathematical Finance, ML in Finance. • Thesis: <i>Kernel Methods on Path Spaces</i> supervised by <i>Prof. Josef Teichmann</i>. BSc Mathematics, University of Pisa 2018-2021 • Honour's degree (110/110 cum Laude) GPA 29.88/30 • Thesis on advanced topics in Geometrical Model Theory.
PUBLICATIONS	[4] Nicola Muca Cirone and Cristopher Salvi. <i>Rough kernel hedging.</i> url: https://arxiv.org/abs/2501.09683 [3] Nicola Muca Cirone, Jad Hamdan and Cristopher Salvi. <i>Graph Expansions of Deep Neural Networks and their Universal Scaling Limits.</i> <i>Submitted to Journal</i> url: https://arxiv.org/abs/2407.08459 [2] Nicola Muca Cirone, Antonio Orvieto, Benjamin Walker, Cristopher Salvi and Terry Lyons. <i>Theoretical Foundations of Deep Selective State-Space Models.</i> <i>Accepted to NeurIPS 2024</i> url: https://arxiv.org/abs/2402.19047 [1] Nicola Muca Cirone, Maud Lemerrier and Cristopher Salvi. <i>Neural signature kernels as infinite-width-depth-limits of controlled ResNets.</i> <i>Proceedings of the 40th International Conference on Machine Learning</i>, 25358–25425, 2023. url: https://proceedings.mlr.press/v202/muca-cirone23a.html
TALKS	Workshop on Recent Developments in Theoretical Machine Learning, Imperial-X Jan 2025 Title: <i>Graph Expansions of Deep Neural Networks.</i> Stochastic Analysis Seminar, ShanghaiTech University Nov 2024 Title: <i>Theoretical Foundations of Deep Selective State-Space Models.</i> DataSig - Rough Paths Interest Group, Alan Turing Institute May 2024 Title: <i>Signature Reconstruction from Randomized Signatures.</i> Computer Vision Group Seminar, University of Michigan April 2024 Title: <i>Theoretical Foundations of Deep Selective State-Space Models.</i> Talks in Financial and Insurance Mathematics, ETH Zürich Dec 2023 Title: <i>Neural Signature Kernels (and Trees).</i>

	7th London-Paris Bachelier Workshop , Imperial College London Title: <i>Rough Kernel Hedging</i> .	Sept 2023
	CDT Mathematics of Random Systems Workshop , University of Oxford Title: <i>Neural Signature Kernels</i> .	June 2023
	16th Oxford-Berlin Young Researchers Meeting , University of Oxford Title: <i>Signatures and the infinite-width-depth limit of Data Driven ResNets</i> .	Dec 2022
EXPERIENCE	Intern - Researcher, Cartesia AI San Francisco, Feb - May 2025 Building new ways to distill LLMs to different architectures, depths and sizes. Research on development of ViT models capable of multi-modal reasoning. Industry Project - Neural Signature Kernels, Imperial College London June 2024 Co-organized 2024 CDT Industry Project and supervised PhD students in applying Neural Signature Kernels to time series analysis. Teaching Assistant - Stochastic Calculus for Finance, ICL Business School Oct - Dec 2023 <i>MSc Risk Management & Financial Engineering</i>. Taught exercise classes. Industry Project - Brain signal analysis with Signatures, Imperial College London June 2023 Challenge: analyze MEG data from several subjects, detect the one who had a seizure. Our team used Signature techniques, the success and clarity of our results won the competition. Teaching Assistant - Numerical Methods for Finance, Imperial College London Feb 2023 <i>MSc Mathematics and Finance</i>. Taught exercise classes. Teaching Assistant - Data Analysis, Università di Pisa 2021-2022 <i>MSc Mathematics</i>. Provided tutoring support. NLP Project - Classification of mathematical papers, Università di Pisa 2021-2022 Task: classify which mathematical subjects an academic paper belongs to, given its title and abstract. The implementation was done in Python.	
AWARDS	NeurIPS 2024 Scholar Award, Neural Information Processing Systems Foundation 2024 INDAM Scholarship, Istituto Nazionale di Alta Matematica 2022 Two year scholarship from the Italian Institute of Higher Mathematics. Placed 2nd at National level.	
COMPETENCES	Languages Italian (<i>native</i>), English (C1), French (<i>basic</i>) Technical Skills Python, Jax, Pytorch, Triton, CUDA, R, Matlab, Java	